

Santiago Guerrero

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EDUCATION

University of San Diego Shiley-Marcos School of Engineering,

Expected May 2027

B.A. Computer Science, Mathematics

GPA: 3.5

Honors: Dean's List, First Honors

SKILLS

Coursework: Data Science, Intro to AI, Advanced Data Structures, Advanced Computational Problem Modeling, Introduction to C++ and Object-Oriented Programming, Programming Abstractions and Methodologies

Programming Languages: Pytorch, Python, FastAPI, Streamlit, Clickhouse, C++, Java, CSS, Javascript, GenAI, SQL, C

Operating Systems: MacOS, Linux

ENGINEERING PROJECTS

Web Design, Personal Project

June 2024 - Present

- Utilized **GenAI**, **JavaScript**, **React**, **Next.js**, and **CSS**. Managed dependencies and configurations using **npm**, **JSON**, and **package management tools** to ensure stability and scalability.
- Developed a working prototype for a categorizing platform specialized in transparent retailer-to-supplier sourcing and communication in the San Diego Used Car Parts industry.

Linked List Parity Project

March 2024 - June 2024

- Constructed data structure with **C++** and **Python** by using nodes which contain references to other nodes, and are designed to expand by iterating through elements in the data set.
- Created a dynamically allocated list object for increased efficiency in data sorting and retrieval.

Wordle Game Project

Fall 2024

- Improved the popular NYT game "Wordle" using **Python**, by utilizing a GUI to enhance user experience, organizing it into modular components, and designed to dynamically update based on player guesses.
- Improved the game by having the boxes of the letters turn green, yellow, or red depending on whether the letter was in the word, in the word but in the wrong place, or not in the word.

PROFESSIONAL EXPERIENCE

USD Office of Undergraduate Research | *Machine Learning Researcher*

June 2025 – Present

- Translated heuristic-based anomaly detection code into a new framework for seamless integration with SLU's **LLM pipeline**, improving drone flight log analysis and supporting a 3–5 member research team.
- Upgraded a heuristic model to an **LSTM autoencoder** to build a new **ML-based anomaly detection pipeline** for drone power logs, achieving a 75% detection rate, a 30% improvement over prior models.

Flo Networks | *Software Engineering Intern*

Summer 2025

- Gained hands-on experience in **network protocols** and infrastructure, working with foundational technologies and conceptual frameworks such as the OSI model, DNS, **IP addressing**, and **routing**.
- Applied these concepts to analyzing data packets, diagnosing network errors, assessing system architecture, and **modeling network topologies**.
- Explored a proprietary system pipeline that uses **FastAPI** to query unstructured data from a database, forward that data to an **LLM** which generates a structured payload, pipe that structured data into **ClickHouse**, and display graphical data using **Streamlit**.

LEADERSHIP EXPERIENCE

Mortar Board National Honor Society | *Co-President*

August 2025 – Present

- Planned and led all the Philanthropic, Social, and Logistical events of the year, including introducing new events and traditions, as well as spearheading the fundraising initiatives and chapter planning.
- Oversee all the committees that oversee individual events: Halloween Hoopla, New Member Selection, etc.

Society of Hispanic Professional Engineers (SHPE) | *Outreach Coordinator*

August 2025 - Present

- Led committee efforts to outreach to K-12 students in order to promote higher learning within marginalized communities through planning the Carson STEAM Night and High Tech High shadow day.
- Outreached to USD organizations to plan collaborative events and promote collaboration within the student body.